

## 1 Pre-Requisites

### 1.1 Distributions

**Binomial**  $X \sim \text{Bin}(n, p)$ :  $M(t) = (pe^t + 1 - p)^n$

**Geometric**  $X \sim \text{Geometric}(p)$ :  $M(t) = \frac{pe^t}{1 - (1-p)e^t}$

**Gamma**  $X \sim \Gamma(\alpha, \lambda)$ :

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}.$$

**Multinomial**  $X \sim \text{Multinomial}(n, \mathbf{p})$ :

$$P(\mathbf{X} = \mathbf{x}) = \binom{n}{x_1, \dots, x_k} \prod_{i=1}^k p_i^{x_i}, \|\mathbf{x}\|_1 = n.$$

$$\text{Cov}(X_i, X_j) = -np_i p_j, \quad i, j \in [1, k], i \neq j.$$

**Hypergeometric**  $X \sim \text{Hypergeometric}(N, m, n)$ :

For  $i \in [\max(0, n + m - N), \min(n, m)]$ ,

$$P(X = x) = \frac{\binom{m}{x} \binom{N-m}{n-x}}{\binom{N}{n}}.$$

$$E(X) = \frac{nm}{N}, \text{Var}(X) = \frac{nm}{N} \frac{N-m}{N} \frac{N-n}{N-1}.$$

### 1.2 Random Variables

#### Expectation and Variance

$$\text{Cov}(X, Y) = E\{(X - \mu_X)(Y - \mu_Y)\}.$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

The **moment generating function** is defined:

$$M_X(t) = E(e^{tX}) = \phi_X(e^t).$$

**Properties of MGF**:

- A MGF uniquely identifies a distribution

- $E[X^k] = \frac{d^k}{dt^k} M_X(t)|_{t=0}$

- $M_{aX+b}(t) = e^{bt} M_X(at)$

The **probability generating function** is defined for discrete RVs:

$$\phi_X(t) = E[t^X] = M_X(\ln t).$$

**Properties of PGF**:

- A PGF uniquely identifies a discrete RV's distribution

- $P(X = 0) = \phi_X(0)$

- $P(X = k) = \frac{1}{k!} \frac{d^k}{dx^k} \phi_X(t)|_{t=0}, \quad k \in \mathbb{N}^*$

**Independence**  $\iff \forall x, y \quad f_{X|Y}(x|y) = f_X(x)$

$$\iff \forall s, t \quad M_{XY}(s, t) = M_X(s) M_Y(t)$$

$$\iff \forall s, t \quad \phi_{X+Y}(s, t) = \phi_X(s) \phi_Y(t)$$

$$\implies E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

$$\implies \text{Cov}(X, Y) = 0$$

**Normal**  $X, Y \wedge \text{Cov}(X, Y) = 0 \implies$  Independence

#### Conditional Distributions

We say  $X_1, \dots, X_n$  are conditionally independent given  $Y$  if

$\forall \langle x_1, \dots, x_n, y \rangle \in \mathbb{R}^{n+1}$ ,

$$P(X_1 = x_1, \dots, X_n = x_n | Y = y)$$

$$= \prod_{i=1}^n P(X_i = x_i | Y = y)$$

$$p_{Y|X}(y|x) = \frac{p_{X|Y}(x|y) p_Y(y)}{\sum_y p_{X|Y}(x|y) p_Y(y)}.$$

With the conditional probability, we may fix  $Y = y$  such that we

have a **conditional RV**  $X|Y = y$ .

$$p_{X|Y=y} = \frac{P(X = x, Y = y)}{P(Y = y)},$$

with **conditional expectation**  $E[X|Y = y]$ .

We may take  $Y$  as the random variable, to obtain the RVs **conditional expectation**  $E[X|Y]$ , and **conditional variance**  $\text{Var}(X|Y)$ .

#### Random Sum

For IID RVs  $X_1, X_2, \dots$ , and discrete RV  $N$  independent of  $X_i$ s, the random sum is defined

$$Y = \sum_{i=1}^N X_i.$$

By the laws of total expectation/variance, we have the results:

$$E[Y] = E(N\mu) = \mu E(N)$$

$$\text{Var}(Y) = E[N\sigma^2] + \text{Var}(N\mu)$$

$$= \sigma^2 E[N] + \mu^2 \text{Var}(N)$$

$$M_Y(t) = E[e^{t \sum_{i=1}^N X_i}] = E[E[e^{t \sum_{i=1}^N X_i} | N]]$$

$$= E[(M_X(t))^N] = E[e^{N \ln M_X(t)}]$$

$$= M_N(\ln[M_X(t)])$$

## 2 Markov Chains

### 2.1 Stochastic Processes

A **stochastic process** is a collection of indexed random variables  $\{X(t, \omega) : t \in T\}$ .

- Given an outcome  $\omega$ ,  $X(\cdot, \omega) : T \rightarrow S$  is a sample path.

- Given an index  $t$ ,  $X(t, \cdot) : \Omega \rightarrow S$  is a distribution on  $S$ .

It is defined with a **state space**  $S$  that contains all possible states, and a **probability measure**  $P$ .

#### Classification

If  $T$  is countable, we have a **discrete-time** stochastic process. Otherwise, it is a **continuous-time** stochastic process.

If  $S$  is countable, we have a **discrete-state** stochastic process. Furthermore if  $S$  is finite, it is a **finite-state** stochastic process. Otherwise, it is a **real-state** stochastic process.

#### Specification

The probability measure for each outcome, of the joint distribution of  $(X_{n_1}, X_{n_2}, \dots, X_{n_k})$  is too difficult to specify in general.

### 2.2 Markov Chain

A stochastic process satisfies the **Markovian property** if given current state  $X_n$ , the next state  $X_{n+1}$  is independent of all past states  $X_{n-1}, \dots, X_0$ .

$$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j | X_n = i).$$

This  $p_{ij}^{n,n+1}$  is known as the **(one-step) transition probability**.

$\{X_n : n \in T\}$  is called a **Markov Chain** if it is a **discrete-time discrete-state stochastic process** satisfying the Markovian property.

#### Specification

A Markov Chain can be **specified** (to find the joint distribution of  $X_n$  at any  $k$  time points) with

- The index set  $T$

- The state space  $S$

- The one-step transition probability  $\forall i, j \in S, n \in T$

We define the **one-step transition probability matrix**  $\mathbf{P}^{n,n+1}$  where each  $\mathbf{P}_{ij}^{n,n+1} = P(X_{n+1} = j | X_n = i)$ . The row vector yields the conditional distribution  $X_{n+1} | X_n = i$ .

#### Chapman-Kolmogorov Equations

For any  $m, n \geq 0$ , we have the equations

$$\mathbf{P}^{n,n+m+1} = \mathbf{P}^{n,n+1} \mathbf{P}^{n+1,n+m+1}$$

$$\mathbf{P}^{n,n+m+1} = \mathbf{P}^{n,n+m} \mathbf{P}^{n+m,n+m+1}.$$

$$p_{ij}^{n,n+m} = \sum_{k_1, k_2, \dots, k_{m-1}} p_{ik_1}^{n,n+1} \dots p_{k_{m-1}j}^{n+m-1,n+m}.$$

### 2.3 Stationary MC

With certain problems, we may further assume that  $p_{ij}^{n,n+1}$  does not depend on  $n$ . We have the **stationary transition probability**

$$p_{ij}^{n,n+1} = p_{ij}, \quad i, j \in S, n \in T.$$

We only require  $S, T, \mathbf{P}$  for the specification of a **stationary MC**.

$$\mathbf{P}^{(m+n)} = \mathbf{P}^{(m)} \mathbf{P}^{(n)} \quad \text{for stationary MCs.}$$

Finite-state stationary MCs can be represented with **graphs**, where nodes represent state, and edges represent possible transitions.

#### Random Walks

Suppose  $\{\xi_n\}_{n \in \mathbb{N}^*}$  a sequence of IID random vectors with  $\xi_i \sim p$ . A stationary MC  $\{X_n\}_{n \in \mathbb{N}}$ , with  $T = \mathbb{N}, S = \mathbb{Z}$  where

$$X_0 = x_0 \in \mathbb{Z}^n, \quad X_{n+1} = X_n + \xi_{n+1}, n \in \mathbb{N}.$$

is known as a **random walk**.

$$X_n = X_0 + \sum_{i=1}^n \xi_i, \quad (\xi_i + 1)/2 \sim \text{Bernoulli}(p).$$

If  $X_0 = 0, (X_n + n)/2 \sim \text{Bin}(n, p)$ .

**Special Cases** of (1-dim) random walks:

- Symmetric/Simple:  $P(\xi_n = 1) = \frac{1}{2}$

- Biased:  $P(\xi_n = 1) = p \neq \frac{1}{2}, \quad P(\xi_n = -1) = q$

- Lazy:  $\xi_n \in \{-1, 1, 0\}, P(\xi_n = 0) = r > 0$

- with Reflecting Boundaries: for  $a, b \in \mathbb{R}, a < b$ ,

$$P(X_{n+1} = a + 1 | X_n = a) = 1$$

$$P(X_{n+1} = b - 1 | X_n = b) = 1.$$

A random walk in  $\mathbb{Z}^d$ , where  $\xi_n$  is a step towards one of its  $2d$  nearest neighbours, is known as a **d-dimensional random walk**.

#### Brownian Motion

Consider a random walk where in each step, the walker moves  $\Delta x \rightarrow 0$  in  $\Delta t \rightarrow 0$  time. This describes a continuous path. If  $\Delta t = c\Delta x^2$  for some constant  $c$ , it is a **Brownian motion**.

### 2.4 Short-Term Analysis

For a MC with state space  $S$ , index set  $\mathbb{N}$  and transition probability matrix  $\mathbf{P}$ , we have the following results:

- $P(X_1 = j | X_0 = i) = p_{ij}$

- Distribution of the **next state**:  $(X_1 | X_0 = i) \sim \mathbf{P}_{i,\cdot}$ .

- Distribution of **n-th state**:  $(X_n | X_0 = i) \sim (\mathbf{P}^n)_{i,\cdot}$ .

- Joint distribution of **two timepoints**  $n \geq m \geq 0$

$$P(X_n = j, X_m = k | X_0 = i) = (\mathbf{P}^{n-m})_{kj} (\mathbf{P}^m)_{ik}$$

- Random initial state**  $X_0 \sim \pi$

$$(X_n = j | X_0 \sim \pi) \sim \pi \mathbf{P}^{(n)}.$$

- Past state conditional** on current and initial state

$$P(X_m | X_n = j, X_0 = i) = \frac{(\mathbf{P}^{n-m})_{kj} (\mathbf{P}^m)_{ik}}{(\mathbf{P}^n)_{ij}}.$$

### 2.5 First Step Analysis

#### Definitions

For strategy-related questions involving the time index  $t$ , we first define a RV  $T : \Omega \rightarrow \{0, 1, 2, \dots, \infty\}$  such that  $\forall n \geq 0$ ,

$\{T = n\}$  depends solely on  $X_1, X_2, \dots, X_n$ ,

$\{T \leq n\}$  depends solely on  $X_1, X_2, \dots, X_n$ .

$T$  is a **stopping time**. Further, a state  $i$  is an **absorbing state** if

$$\forall j \in S - \{i\}, p_{ij} = 0$$

#### First Step Analysis

Using First Step Analysis, we may solve **problems related to stopping time**  $T$ , such as the state at time  $T$ ,  $E[T]$ , functions of  $T$ .

Consider  $a_i(T) = h(i, \dots, X_T | X_0 = i)$ :

- Consider the first step, where

$$a_i(T) = \sum_{k \in S} h(i, k, X_2, \dots, X_T | X_0 = i, X_1 = k) P(X_1 = k | X_0 = i).$$

- Consider the probabilistic equivalent process  $\{Y_n\}$ .

$$h(i, k, X_2, \dots, X_T | X_0 = i, X_1 = k)$$

$$= g(h(k, X_1, \dots, X_T | X_0 = k)).$$

where  $g$  depends on the specified problem.

- Define  $a_i(T) = h(i, \dots, X_T | X_0 = i)$  for each state  $i \in S$ , and express the equation as

$$a_i(T) = \sum_{k \in S} g(a_k(T)) P(X_1 = k | X_0 = i).$$

- Solving the system of equations of  $a_i, \forall i \in S$ , we obtain the desired result.

#### Strategy-Related Problems

To obtain  $P(X_T = S_A | X_0 = S_i)$ , we take the first step of the MC with the Chapman-Kolmogorov equations. Consider  $Y_n = X_{n+1}$ .  $\{Y_n\}$  has the **same probabilistic structure** as  $\{X_n\}$ .

$$P(X_T = A | X_1 = i) = P(Y_{T-1} = 0 | Y_0 = i).$$

$T_Y := \min\{n : Y_n = S_A\}$ .  $T_Y$  is a stopping time of  $\{Y_n\}$ .

$$P(Y_{T_Y} = 0 | Y_0 = i) = P(X_T = 0 | X_0 = i).$$

$$\therefore P(X_T = 0 | X_1 = 1) = P(Y_{T-1} = 0 | Y_0 = 1)$$

$$= P(Y_{T_Y} = 0 | Y_0 = 1) = P(X_T = 0 | X_0 = 1)$$

Defining  $u_i = P(X_T = S_A | X_0 = S_i)$ , we have a system of equations in  $u_i, \forall i \in S$ .

#### General Gambler's Ruin

Suppose a gambler who will stop with  $N$ , or 0 dollars in hand. In each game, they win with probability  $p$ , and lose with  $q = 1 - p$ . Beginning with  $0 < k < N$  in hand, the **probability of stopping at 0**  $u_k = P(X_T = 0 | X_0 = k)$  is

$$\begin{cases} u_k = 1 - \frac{k}{N} & p = q \\ u_k = 1 - \frac{1 - (q/p)^k}{1 - (q/p)^N} & p \neq q \end{cases}$$

The **expected number of games** played  $v_k = E[T | X_0 = k]$  is

$$\begin{cases} v_k = k(N - k) & p = q \\ v_k = \frac{1}{p - q} \left[ \frac{N(1 - (q/p)^k)}{1 - (q/p)^N} - k \right] & p \neq q \end{cases}$$

## 3 Long Run Performance

### 3.1 Classification of States

#### Communicating Classes

State  $j$  is **accessible from state**  $i$ ,  $i \rightarrow j$  if

$$\mathbf{P}_{ij}^m > 0 \text{ for some } m \geq 0.$$

If  $i \rightarrow j \wedge j \rightarrow i$ , they are **communicating**,  $i \leftrightarrow j$ .

Communication is an **equivalence relation**. We define  $(S / \leftrightarrow)$  the set of communication classes, each denoted  $\mathcal{C}$ . It is **well-defined**.

Classes do not communicate. If  $i \in \mathcal{C}_1, j \in \mathcal{C}_2$  such that  $i \rightarrow j$ ,  $\forall i' \in \mathcal{C}_1, j' \in \mathcal{C}_2 \quad i' \rightarrow j', \quad j \not\rightarrow i, \quad j' \not\rightarrow i'$ .

A MC is **irreducible** if there is only one communication class.

#### Recurrence

For state  $i$ , the **first return probability** is the probability of having the  $n$ -th transition be the first return to  $i$  with initial state  $i$

$$f_{ii}^{(0)} := 0.$$

$$f_{ii}^{(n)} = P(X_1 \neq i, \dots, X_{n-1} \neq i, X_n = i | X_0 = i).$$

The probability of **revisiting**  $i$  in the future is

$$f_{ii} = \sum_{n=0}^{\infty} f_{ii}^{(n)} \leq 1,$$

and the **return probability** to  $i$  from  $i$  at the  $n$ -th transition is

$$\mathbf{P}_{ii}^{(n)} = P(X_n = i | X_0 = i).$$

We have the **convolution**

$$P_{ii}^{(n)} = \sum_{k=1}^n f_{ii}^{(k)} P_{ii}^{(n-k)}.$$

The **expected number of revisits** to state  $i$ ,  $E[N_i | X_0 = i]$  is

$$E[N_i | X_0 = i] = \sum_{n=1}^{\infty} P_{ii}^{(n)} = \begin{cases} \infty & f_{ii} = 1 \\ \frac{f_{ii}}{1-f_{ii}} & f_{ii} < 1 \end{cases}$$

$$\begin{aligned} i \text{ is } \textbf{recurrent/not recurrent} &\iff f_{ii} = 1 \\ &\iff E[N_i | X_0 = i] \text{ is infinite} \\ &\iff \sum_{n=1}^{\infty} P_{ii}^{(n)} \end{aligned}$$

Recurrence is a **class property**.

**Periodicity**

For a state  $i$ , we define the **period** of  $i$

$$d(i) := \begin{cases} \gcd R_{ii} & R_{ii} \neq 0 \\ 0 & R_{ii} = 0 \end{cases}$$

where  $R_{ij} := \{n \geq 1 : P_{ij}^{(n)} > 0\}$ .

$$\begin{aligned} &\bullet \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, 0 < k < d(i) \\ &\quad \mathbf{P}_{ii}^{(n+d(i))} > 0, \quad \mathbf{P}_{ii}^{(n+d(i)+k)} = 0. \\ &\bullet \forall m \in R_{ji}, \text{ for the same } N, \forall n \geq N \\ &\quad \mathbf{P}_{ji}^{(m+nd(i))} > 0. \end{aligned}$$

Periodicity is a **class property**. A class with period 1 is **aperiodic**.

**Regularity and Ergodicity**

A MC is **regular** if  $\exists k \in \mathbb{N}$  such that

$$\forall i, j \in S, \quad (\mathbf{P}^{(k)})_{ij} > 0.$$

$\mathbf{P}$  of a regular MC is a regular transition probability matrix.

### 3.2 Recurrent Irreducible MC

The **expected duration between revisits** is defined

$$m_i = E[R_i | X_0 = i] = \sum_{n=1}^{\infty} n f_{ii}^{(n)}.$$

If  $m_i$  is finite, the MC is **positive recurrent**. Otherwise, it is null recurrent. Given a fixed path, consider the **long-run proportion of visits**. It is given by  $\lim_{n \rightarrow \infty} E\left[\frac{1}{n} \sum_{k=1}^n I(X_k = j | X_0 = i)\right]$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \mathbf{P}_{ij}^{(k)} = \frac{1}{m_j}.$$

If the MC is periodic:  $\lim_{n \rightarrow \infty} \mathbf{P}_{jj}^{(nd)} = \frac{d}{m_j}$ .

If the MC is aperiodic and null recurrent:  $\lim_{n \rightarrow \infty} \mathbf{P}_{ij}^{(n)} = 0$ .

**Finite MC:** Regular  $\iff$  Ergodic  $\iff$  Irreducible, Aperiodic.

**Infinite MC:** Ergodic  $\iff$  Irreducible, Aperiodic, + Recurrent.

**Ergodic MC**

$\lim_{n \rightarrow \infty} \mathbf{P}_{ij}^{(n)}$  exists independent of initial state  $i$ .

$$\lim_{n \rightarrow \infty} \mathbf{P}_{ij}^{(n)} = \lim_{n \rightarrow \infty} \mathbf{P}_{jj}^{(n)} = \frac{1}{m_j} = \pi_j.$$

Since  $\sum_{k=1}^N \pi_k = 1$ , we define the **limiting distribution**

$$\pi := \langle \pi_1, \pi_2, \dots, \pi_N \rangle.$$

$\pi$  is obtained from the solution to  $\pi \mathbf{P} = \pi$ ,  $\|\pi\|_1 = 1$

$$\begin{cases} \pi_j = \sum_{k=1}^N \pi_k \mathbf{P}_{kj} & j \in \{1, 2, \dots, N\} \\ \|\pi\|_1 = 1 \end{cases}$$

The regular transition probability matrix converges

$$\lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} \pi \\ \pi \\ \vdots \\ \pi \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ \vdots \\ N \end{pmatrix} \begin{pmatrix} \pi_1 & \pi_2 & \dots & \pi_N \\ \pi_1 & \pi_2 & \dots & \pi_N \\ \vdots & \vdots & \ddots & \vdots \\ \pi_1 & \pi_2 & \dots & \pi_N \end{pmatrix}$$

That is,  $X_n \sim \pi^{(0)} \mathbf{P}^{(n)} \rightarrow \pi$  for any initial distribution.

Given a large time point  $n$ ,  $X_n$  is a **random variable**. Each path gives a realisation of  $X_n$  and  $P(X_n = k) = \pi_j$ .

Given a path, consider the **proportion of visits** over time.

$E\left[\frac{1}{n} \sum_{k=0}^{n-1} I(X_k = j | X_0 = i)\right] \rightarrow \pi_j$  as  $n \rightarrow \infty$ .

The **mean duration between visits** to  $j$  is given by  $\frac{1}{m_j}$ .

### 3.3 General MC Long Run Performance

1. Obtain the set of communicating classes  $\mathcal{C}_k$
2. Set up a MC such that each recurrent class is denoted by a node
3. Define  $T := \min\{n : X_n \in \text{recurrent classes}\}$
4. Using FSA, find  $u_{k|i} = P(X_T \in \mathcal{C}_k | X_0 = i)$ .
5. Within each recurrent class  $\mathcal{C}_k$ :
  - Period  $> 1$ : Long-run proportion of time in each state.
  - Period  $= 1$ : If finite or infinite positive recurrent, find the corresponding limiting distribution of state  $j \in \mathcal{C}_k$ , denoted  $\pi_{j|k}$ . Otherwise, the probability at each state is 0.

6. For any initial state  $X_0 = i$ :

- If  $j$  is transient,  $\pi_{j|i} = 0$
- If  $j \in \mathcal{C}_k$ ,  $\pi_{j|i} = u_{k|i} \pi_{j|k}$

7. With an initial distribution  $X_0 \sim \pi^{(0)}$

$$\pi_{j|\pi^{(0)}} = \sum_{i \in S} \pi_{j|i} \pi_i^{(0)}.$$

### 3.4 Balance Equations

The **global balance equations**  $\pi = \pi \mathbf{P}$  yields the limiting distribution. Equivalently, we have the  $N$  equations:

$$\pi_j = \sum_{k \in S} \pi_k \mathbf{P}_{kj}.$$

$\pi$  exists even for non-ergodic MCs:  $(X_{n+1} | X_n \sim \pi) \sim \pi$ .

|                        | Limiting Dist.                                                          | Stationary Dist.                 |
|------------------------|-------------------------------------------------------------------------|----------------------------------|
| Initial dist.          | $X_0 \sim \pi^{(0)}$                                                    | $X_0 \sim \pi$                   |
| First step             | $X_1 \sim \pi^{(0)} \mathbf{P}$                                         | $X_1 \sim \pi \mathbf{P} = \pi$  |
| $n \rightarrow \infty$ | $X_n \sim \pi$                                                          | $X_n \sim \pi$                   |
| Calculation            | $\pi = \pi \mathbf{P}$<br>or $\lim_{n \rightarrow \infty} \mathbf{P}^n$ | $\pi = \pi \mathbf{P}$           |
| Uniqueness             | Yes                                                                     | No                               |
|                        | $\pi_1 = \pi_2$                                                         | it's possible $\pi_1 \neq \pi_2$ |
| MC                     | Ergodic                                                                 | No requirement                   |

A **unique** stationary distribution is the limiting distribution.

Consider a transition matrix  $\mathbf{M}$  such that

$$\mathbf{M}_{ij} = P(X_{n+1} = j, X_n = i) = \pi_i \mathbf{P}_{ij},$$

The row sum equals the column sum for any  $i, j \in S$ . This can be obtained with the  $\binom{n}{2}$  **local (detailed) balance equations**. Suppose there exists a distribution  $\pi$  such that

$$\pi_i \mathbf{P}_{ij} = \pi_j \mathbf{P}_{ji}.$$

Then matrix  $\mathbf{M}$  is symmetric.  $\pi$  also satisfies the **global balance equations**, and is easier to compute.

## 4 Applications of MC

### 4.1 PageRank

Given a series of webpages  $S$ , each having  $k_i$  hyperlinks to other webpages, consider a random walk on the stationary MC:

• State Space  $S$ , Time index  $\mathbb{N}$

$$P_{ij} = \begin{cases} \frac{1}{k_i} & \text{if there is a hyperlink } i \rightarrow j \\ 0 & \text{otherwise} \end{cases}$$

$$\pi^{(0)} = \left(\frac{1}{|S|}, \dots, \frac{1}{|S|}\right)$$

If the MC is nonregular, consider instead

$$\tilde{\mathbf{P}} = \lambda \mathbf{P} + (1 - \lambda) \pi_0 \mathbf{1}^T$$

$$\pi_{n+1} = \lambda \pi_n \mathbf{P} + (1 - \lambda) \pi_0.$$

for some damping factor  $\lambda \in (0, 1)$ . The limiting distribution  $\pi$  is used to rank in order of decreasing limiting probability.

### 4.2 Markov Chain Monte Carlo

Given a target distribution  $\mathbf{p} = c_n \mathbf{p}'$ , we find a MC such that  $\pi = \mathbf{p}$ , using the **Metropolis-Hastings Algorithm**:

1. Initialise with any initial state  $X_0$ , and an ergodic MC with the same sample space as  $\mathbf{p}$  and transition probability matrix  $\mathbf{Q}$

$$\mathbf{Q}_{ij} = P(Y_{n+1} = j | Y_n = i)$$

2. Using  $\mathbf{Q}_{X_n}$ , draw a proposal state  $Y$ .

3. Calculate the acceptance rate of the proposal state

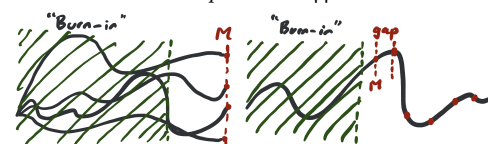
$$\alpha(X_n, Y) = \min\left(\frac{p'_Y}{p'_{X_n}} \cdot \frac{Q_{Y, X_n}}{Q_{X_n, Y}}, 1\right).$$

4. If the proposal state is accepted,  $X_{n+1} = Y$ .

On a **continuous** distribution  $f = c_n f'$ , the state space  $S$  is a continuous set, the transition probability is the **density function**  $g_X(y) = g(y|x)$  and the **acceptance rate** is the analogue

$$\min\left(\frac{f'(Y)}{f'(X_n)} \cdot \frac{g_Y(X_n)}{g_{X_n}(Y)}, 1\right)$$

For sufficiently large  $M$  and  $gap$ , we can simulate sampling from  $p$  from the MC to simulate  $p$  with two approaches.



### 4.3 Branching Process

A **branching process** is a stochastic process  $\{X_n\}_{n=0}^{\infty}$  where, for some IID RVs  $\xi_i^{(j)}$  over the sample space  $\mathbb{N}$ ,

$$X_{n+1} = \sum_{i=0}^{X_n} \xi_i^{(n)}.$$

**Properties of Branching Processes:** A **stationary MC** of known transition probability function, **aperiodic** with two state classes:  $\{0\}$  and  $\mathbb{N}^*$ .

$$\phi_{X_{n+1}|X_n}(t) = \phi_{\xi}^{X_n}(t)$$

$$\phi_{X_{n+1}}(t) = E\left[\phi_{\xi}^{X_n}(t)\right] = \phi_{X_n}(\phi_{\xi}(t))$$

$$\text{If } X_0 = k, \phi_{X_n}(t) = \left[\phi_{\xi}^{(n)}(t)\right]^k.$$

With the known **moments of  $\xi$** ,

$$E[X_n] = \mu E[X_{n-1}] = \mu^n.$$

$$Var(X_n) = \mu^{n-1} \sigma^2 \times \begin{cases} \frac{1-\mu^n}{1-\mu} & \mu \neq 1 \\ n & \mu = 1 \end{cases}.$$

## Extinction Probability

For absorbing state 0, we define the stopping time and denote

$$T = \min\{n : X_n = 0\}, \quad u_n^{(k)} := P(T \leq n | X_0 = k)$$

Since  $\{T \leq n\} = \cap_{i=1}^n \{T \leq i\} \implies u_n^{(k)} = u_n^{(k)}$ ,

$$u_n = \sum_{l \in S} P(\xi = l) u_{n-1}^l = \phi_{\xi}(u_{n-1}),$$

and  $u_1 = P(\xi = 0)$ , we can find the **extinction probability before generation  $n$**   $u_n^{(k)}$  by

$$u_n^{(k)} = u_n^k = \left[\phi_{\xi}^{(n-1)}(P(\xi = 0))\right]^k.$$

The eventual **extinction probability** exists as it is a non-decreasing probability wrt.  $n$ .  $\phi_{\xi}(0) = P(\xi = 0)$ ,  $\phi_{\xi}(1) = 1$ , and it is a **convex increasing function** wrt  $x$ , we have the classes of solutions:

- $P(\xi = 0) = 0 \implies u_{\infty} = 0$
  - $P(\xi = 0) > 0, E[\xi] < 1 \implies u_{\infty} = 1$ . **Subcritical**.
  - $P(\xi = 0) > 0, E[\xi] = 1 \implies u_{\infty} = 1$ . **Critical**.
  - $P(\xi = 0) > 0, E[\xi] > 1 \implies u_{\infty} < 1$ . **Supercritical**.
- $u_{\infty}$  is the solution to  $x = \phi_{\xi}(x)$ .

### 5 Poisson Process

For independent events over  $[0, t_0]$  decomposed into  $n \rightarrow \infty$  intervals of length  $h = \frac{t_0}{n}$ ,  $N((kh, kh+h]) \sim \text{Bern}(p_n)$  and  $N([0, t_0]) \sim \text{Bin}(n, p_n)$ . If  $p_n \in o(h)$ ,  $np_n \rightarrow \lambda$ ,

$$\begin{aligned} P(X = x) &= \binom{n}{x} \frac{\lambda^x}{n} \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ &= \frac{n!}{n^x} \frac{\lambda^x}{x!} \left(1 - \frac{\lambda}{n}\right)^{n-x} \rightarrow \frac{\lambda^x}{x!} e^{-\lambda} \end{aligned}$$

We have the **Poisson Distribution**  $X \sim \text{Pois}(\lambda)$  with  $\lambda > 0$ ,

$$P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x \in \mathbb{N}.$$

$$E[X] = Var(X) = \lambda, \quad M_X(t) = e^{\lambda(e^t - 1)}.$$

A **Poisson process of rate/intensity**  $\lambda > 0$  is a continuous-time, discrete-state stochastic process  $\{X(t) | t \geq 0\}$  where

- $X(t_1) - X(t_0), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$  are independent RVs, for any  $t_0 = 0 < t_1 < \dots < t_n$
- $X(s+t) - X(s) \sim \text{Pois}(\lambda t)$ ,  $s \geq 0, t > 0$
- $X(0) = 0$

Alternatively, by the **law of rare events**,  $N((s, t])$  is a **Poisson process** of intensity  $\lambda > 0$  if

- $N((t_0, t_1]), \dots, N((t_{n-1}, t_n])$
- $\exists \lambda, \quad P(N((t, t+h]) \geq 1) = \lambda h + o(h), \quad h \rightarrow 0$
- $\exists \lambda, \quad P(N((t, t+h]) \geq 2) = o(h), \quad h \rightarrow 0$

This allows generalisation:  $\lambda \rightarrow \lambda(t)$  yields a **non-homogeneous Poisson process**,  $h \rightarrow \text{Vol}(A)$  a **Poisson point process**.

For a Poisson process  $X(t)$  of rate  $\lambda$ , the **waiting time** of the  $n$ -th event  $W_n \sim \text{Exp}(\lambda) = \Gamma(n, \lambda)$ . Further, the **sojourn times**  $S_n = (W_{n+1} - W_n) \sim \text{Exp}(\lambda)$ .

This gives the alternative definition: the counting process with waiting times  $W_n = S_0 + S_1 + \dots + S_{n-1}$  is a **Poisson process** of rate  $\lambda$  if

$$\{S_n : n \geq 0\} \text{ are IID RVs, } S \sim \text{Exp}(\lambda)$$

With the memoryless property, given that  $X(t) = n$ , each of the events occur at any  $t \in [0, t_0]$  with equal probability. That is, The **conditional distribution** of  $W_1, W_2, \dots, W_n$ , given  $X(t) = n$ , is the same as the **order statistics** corresponding to  $n$  IID uniform RVs over  $(0, t)$ .

$$f_{W_k}(x) = \frac{n!}{(n-k)!(k-1)!} \frac{1}{t} \left(\frac{x}{t}\right)^{k-1} \left(\frac{t-x}{t}\right)^{n-k}$$